



Sproutfit: an immersive seed-planting virtual reality game to enhance patient motivation for performing exercises for the prevention of venous thromboembolism through loss and avoidance gamification

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Abstract

Sustaining patient engagement in physical rehabilitation for conditions of venous thromboembolism (VTE) prevention is often challenging due to factors such as the repetitive nature of the rehabilitation exercises, and possible insufficient patient motivation. This study investigated the integration of immersive virtual reality (IVR) and gamification to address this challenge. This study explored whether or not integrating the loss and avoidance (LA) gamification dynamic into an IVR rehabilitation game can improve short-term motivation to perform VTE-prevention exercises among middle-aged and older hospital patients. A mixed-design experiment was carried out that involved 28 patients admitted for post-surgical rehabilitation. A smart wearable sensor-based IVR game (SPROUTFIT) was developed, featuring a theme of seed-planting in five stages, where patients perform lower limb exercises to simulate gardening activities. The game employs a level system (“Novice Planter”, “Adept Planter” and, “Master Planter”) to reward performance, with the “loss” component creating a fear of not retaining levels, and the “avoidance” component requiring patients to repeat stages if performance criteria are not met. The patients in the LA-integrated SPROUTFIT (LA-SPROUTFIT) showed significantly higher external regulation compared to the non-LA-integrated SPROUTFIT (NLA-SPROUTFIT) group. LA-SPROUTFIT group also demonstrated significantly higher scores regarding the game experience of competence, flow, and challenge. Game scores were significantly higher in LA-SPROUTFIT group for game stages 2, 4, and 5. Physiological measures indicated that LA-SPROUTFIT resulted in significantly higher heart rate, calorie expenditure, and perceived exertion. Qualitative feedback from interviews revealed that patients generally perceived SPROUTFIT as enjoyable and motivating. However, some patients suggested including various difficulty levels and additional personalization options in future iterations.

Keywords Loss and avoidance · Rehabilitation · Immersive virtual reality · Gamification

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Abbreviations

IVR	Immersive virtual reality
VTE	Venous thromboembolism
HMD	Head-mounted display
LA	Loss and avoidance
NLA	Non-loss and avoidance
RQ	Research question
MAPs	Middle-aged patients
OPs	Older patients
GC	Game condition
HR	Heart rate
IMU	Inertial measurement unit
GAMS	Gaming motivation scale
GEQ	Game experience questionnaire
Borg RPE	Borg rating of perceived exertion

1 Introduction

Venous thromboembolism (VTE), which includes both deep vein thrombosis and pulmonary embolism, is a chronic disease that affects approximately 10 million people worldwide each year (Khan et al. 2021). VTE survivors often experience varying levels of functional impairments, such as post-thrombotic syndrome and pulmonary hypertension, which require ongoing management or prevention (Lutsey et al. 2020). To address these post-VTE issues, rehabilitation is strongly recommended as a preventive measure against VTE (Noack et al. 2015). Among the most commonly prescribed rehabilitation strategies is exercise (Kahn et al. 2008), which has been shown to aid recovery and reduce the likelihood of future VTE incidents (Rolving et al. 2020; Xu et al. 2021). Studies have shown that exercise does not negatively impact blood clot and may decrease the risk of post-thrombotic syndrome (Kahn et al. 2003; Lakoski et al. 2015). However, the lack of standardized guidelines for rehabilitation exercises aimed at preventing VTE has led to inconsistent instructions and decreased adherence to exercise regimens (Rolving et al. 2020). This inconsistency, coupled with the diversity of patient needs, makes the development of standardized rehabilitation protocols challenging for clinicians (Kelly et al. 2010).

One promising solution to these challenges lies in the use of virtual reality (VR), particularly immersive VR (IVR) technology that uses head-mounted displays (HMDs). IVR has recently gained popularity as a promising approach in healthcare, with rapid incorporation into clinical settings (Laver et al. 2017; Pillai and Mathew 2019). IVR has the potential to transform rehabilitation by offering personalized and interactive experiences for patients with physical or cognitive impairment (Patsaki et al. 2022; DiCesare et al. 2019; Punt et al. 2017). Despite the advantages of IVR,

maintaining patient engagement and motivation throughout the rehabilitation process remained a challenge: Integrating gamification strategies within IVR may increase patient adherence and motivation.

Gamification, defined as the integration of game-design elements into non-game contexts (Deterding et al. 2011), aims to enhance user engagement and motivation (Seaborn and Fels 2015). Werbach and Hunter's framework provides a hierarchical structure for categorizing game-design elements, distinguishing between dynamics, mechanics, and components (Werbach and Hunter 2015). Typically, gamification implementations integrate game mechanics (such as reward systems) with tangible game components (such as achievements and levels) to give concrete or symbolic recognition to users for completing tasks (Sardi et al. 2017). Social influence, frequently facilitated through competitive or collaborative structures, can be another game mechanic that enhances user engagement (Werbach and Hunter 2015). This is particularly important given that social influence resonates with the fundamental psychological needs of relatedness and belonging, which are effective drivers of motivation according to self-determination theory (Deci and Ryan 2008). Within the healthcare sector, the application of game dynamics, such as narrative, along with game components that include avatar-based self-representation, contribute to the effectiveness of gamified interventions (Damaševičius et al. 2023; Miller et al. 2016).

The Octalysis framework (Chou 2019), a gamification and behavior-design model, offers a structured approach to understand and apply human motivational principles in non-game environments. Previous studies across diverse domains has shown its ability to influence user engagement and behavior (Mohanty and Christopher B 2023; Chen et al. 2023; Xu and Fang 2023; Heryanto et al. 2024). Comprised of eight core motivational drives, the framework facilitates the creation of more engaging and compelling experiences. Among these drives, Loss & Avoidance (LA) is notable for utilizing the fear of losing something important or facing adverse outcomes (Chou 2019). From a neurobiological perspective on motivation, this drive is associated with the brain's sensitivity to potential loss: It involves regions of the brain that are linked to processing negative emotions, and behavior aimed at avoiding aversive stimuli (Hayes and Northoff 2011). In gaming contexts, setbacks, such as losing coins, lives, or progress, are commonly used to motivate players to avoid losses and continue to advance (Engelstein 2020).

The impact of the LA drive is subject to individual variability (Chou 2019; Boyce et al. 2016). Although the prospect of loss may be a powerful motivator, promoting participation and adherence, it may also produce anxiety or frustration in some people (Rewley et al. 2021). These

different responses to LA are influenced by multiple factors, including personality traits, previous experiences, and perceptions of control: The LA drive, therefore, requires careful consideration, particularly for healthcare and rehabilitation applications. Furthermore, in line with the existing literature on gamified rehabilitation exercises, which demonstrates a varied focus and often does not consistently prioritize gender and player types as primary variables (Garcia-Hernandez et al. 2024; Dalko et al. 2025; Shah et al. 2023; Alex et al. 2021; Zhang et al. 2024; Howard 2017; Tao et al. 2021), our study also did not designate these factors as primary variables for investigation.

To address the low adherence often observed in VTE prevention exercises for bedridden patients, and to improve active participation during prescribed sessions, we strategically integrated the LA drive into our rehabilitation game design. This approach aligns with established psychological principles regarding loss aversion (Engelstein 2020). The time-limited and performance-based structure inherent in the IVR game facilitates the immediate perception of potential progress loss, making LA a particularly relevant and potent motivator in this context. Despite its potential effectiveness in gaming, the application of LA in rehabilitation settings remains largely unexplored. Investigating how LA can be adapted to promote sustained participation, and to improve rehabilitation outcomes, represents a promising avenue for enhancing patient engagement in healthcare.

The key contributions of this research are as follows:

- We introduce SPROUTFIT, an IVR-rehabilitation seed-planting game, specifically designed for the prevention of VTE. SPROUTFIT integrates the LA gamification drive, aiming to significantly improve the overall rehabilitation experience by creating a more engaging, entertaining and motivational environment for patients, ultimately promoting greater adherence to rehabilitation protocols (Maynard and Stein 2010; Noack et al. 2015).
- We provide a comprehensive quantitative and qualitative analysis of the effects of LA and non-LA (NLA) on patient motivation, gaming experience, gaming performance, and exertion in the (middle-aged (MAPs) and older (OPs) patients).

2 Related work

In rehabilitation, IVR interventions have been used in various contexts, such as musculoskeletal rehabilitation (Schuermans et al. 2022), neurological recovery (Patsaki et al. 2022), cognitive therapy (Chatterjee et al. 2022), and psychological treatment (Gatto et al. 2020). These applications have addressed multiple conditions, including stroke

(Dong et al. 2023), spinal cord injury (Solcà et al. 2021), amputations (Lancere et al. 2023), brain injuries (Glegg et al. 2017), and mental health disorders (Pons et al. 2022). This use of IVR has shown great potential to improve therapeutic outcomes.

The benefits of IVR in rehabilitation settings has also been explored. For example, Parente et al. (2023) designed an IVR system for upper limb rehabilitation, integrating biofeedback to adjust difficulty based on patient's heart rate (HR): They found that patient engagement increased, potentially improving rehabilitation outcomes. Sun et al. (2023) introduced an IVR application using musical instruments to motivate stroke patients through multi-sensory feedback. Villiger et al. (2017) developed different IVR scenarios that target lower limb muscle training: These improved muscle strength, balance, and functional mobility. Systems like these allow patients to practice their movements in a safe and controlled environment, enhancing their sense of agency and embodiment (Baltaci et al. 2013; Ratcliffe et al. 2021). Ziadeh et al. (2021) studied the impact of the IVR embodiment on agency and frustration during motor-imagery tasks; Matamala-Gomez et al. (2022) examined their impact on accelerating motor recovery during upper limb rehabilitation. Other researchers, such as Izountar et al. (2022) and Lai et al. (2018), developed adaptive IVR systems that included patient emotions and real-time data to personalize rehabilitation and improve engagement. Despite these advancements, the use of IVR in preventing VTE remains underexplored. Rehabilitation regarding VTE prevention requires precise movements of the lower limbs (Wang et al. 2016), while current IVR technologies often struggle to accurately capture. This presents a critical challenge that warrants further investigation to address the unique needs of VTE prevention.

Gamification has gained considerable attention in rehabilitation: It aims to increase motivation through the integration of game elements into therapeutic exercises (Seaborn and Fels 2015). Zichermann (2011) classified motivation in gamification into intrinsic and extrinsic forms; Intrinsic motivation arises from the inherent satisfaction derived from an activity, such as creativity or social interaction, while extrinsic motivation arises from the pursuit of external rewards (Chou 2019). Game dynamics such as narrative and storytelling enhance intrinsic motivation by transforming routine exercises into exciting adventurous stories (Kern et al. 2019). Eckert et al. (2022) reported that patients may find motivation through narrative and high score achievement in a modular exercise design. Similarly, Elor and Kurniawan (2020) integrated narrative into an IVR game, fostering a sense of connection and responsibility. Birk and Mandryk (2019) showed that avatar customization is

also a key factor for increasing engagement in gamified rehabilitation.

In contrast, extrinsic motivators (such as scores, levels, and leaderboards) have been widely used to encourage patients to achieve specific goals (Kern et al. 2019; Tuah et al. 2021). For example, Menychtas et al. (2022) provided instant feedback through rewards (in-game scores) and comparisons (performance comparisons with other users through leaderboards) in the study that showed an increase in patient motivation. A combination of intrinsic and extrinsic motivators has proven essential to maintain long-term involvement in rehabilitation (Muangsriinon and Boonbrahm 2019; Richter et al. 2015). An effective gamified rehabilitation design requires careful consideration of movement-based guidelines: Isbister et al. (2011) recommended embracing ambiguity in movement-based games, turning variability into opportunities for patient self-exploration. Lazzaro (2004) highlighted that improving movement quality and managing cognitive load can be key strategies to maintain accessibility and enjoyment. In addition, Karageorghis and Priest (2008) showed that the inclusion of rhythm and intentional fatigue mechanics (game designs that strategically induce a manageable level of exertion or challenge) further supported patient engagement and progression.

Grounded in the established psychological principle of loss aversion (Engelstein 2020), LA is recognized as a core motivational driver within gamification frameworks (Chou 2019). The LA principle highlights an asymmetry in psychological impact: The prospect of incurring a loss or facing an adverse outcome exerts a more powerful motivational force than the potential for an equivalent gain. In the design of gamified systems, LA is strategically implemented to take advantage of this human tendency by framing continued engagement as a means of preventing negative consequences (such as the forfeiture of accumulated progress, status, virtual rewards or opportunities). The application of LA is evident across various domains. In educational contexts, for instance, LA has been used to promote consistent learning behaviors. Finardi et al. (2016) refers to the concept of a “learning streak” in mobile language learning applications, where the risk of losing the streak incentivizes regular practice. Similarly, Plangger et al. (2022) emphasized that, with fitness applications, LA has motivated sustained participation, often by highlighting the potential loss of achieved progress if engagement stops. The psychological phenomenon of the endowment effect suggests that individuals ascribe greater value to items they already possess, thus increasing the importance of the threat of losing them (Drouvelis and Sonnemans 2017). This effect can be strategically used in gamification by giving users virtual possessions, or by acknowledging achievements early in their

interaction, consequently increasing their motivation to avoid losing them.

Although previous studies have shown progress in IVR and gamified rehabilitation, certain challenges persist. The integration of LA with IVR and gamified elements, particularly for lower limb exercises aimed at preventing VTE, remains underexplored. There is also a lack of comprehensive understanding on how individual psychological factors can moderate the effectiveness and acceptance of IVR interventions. Addressing the issues regarding these individual differences is crucial for the development of truly personalized and effective rehabilitation strategies. The current study sought to use LA-SPROUTFIT to enhance patient adherence to lower limb rehabilitation tasks. Furthermore, while many studies have examined intrinsic and extrinsic motivators in such a context (Zhang and Yu 2024), more studies are required to determine how age-related factors impact the efficacy of these motivators. Yen and Chiu (2021) reported that IVR-based exergames (exercise video games) have achieved acceptance rates (the percentage of participants who agreed to continue using the intervention) ranging from 80% to 98% among older adults, highlighting their potential to promote active aging and support mental health. Similarly, Dermody et al. (2020) indicated the potential of IVR to address physical and cognitive challenges in older adults. They also emphasized the need to explore additional factors that could enhance the effectiveness of IVR for such patients. Furthermore, when designing and implementing these technologies, it is necessary to consider not only motivational elements, but also potential ethical implications and accessibility requirements. Ethical considerations include informed consent processes, the protection of patient privacy, and the minimization of possible adverse effects (such as discomfort or disorientation) (Brown et al. 2022). Addressing accessibility challenges involves accommodating the diverse physical limitations, cognitive impairments, and sensory deficits prevalent among older adults: This requires a careful and inclusive application design (Tuena et al. 2020).

In summary, the literature review highlights the significant potential of IVR and gamification in diverse rehabilitation domains. However, the specific application of the LA motivational driver within this context for VTE prevention through lower limb exercises remains notably underexplored. Although previous research provides valuable information on IVR-based movement therapy and the motivational efficacy of gamification, a comprehensive understanding of how individual psychological factors and age-related differences influence responses to these approaches is still lacking. In addition, challenges remain for accurately capturing intricate lower limb movements using current IVR technologies. There is also a need to

address ethical considerations and ensure accessibility for heterogeneous patient populations. Consequently, more research is essential, to explore the integration of LA within IVR environments for VTE-prevention rehabilitation, with a focus on addressing these identified gaps in the current literature.

To address these gaps, the following research questions (RQs) were formulated:

- *RQ1: How do LA dynamics influence patient short-term motivation and the perceived quality of the IVR-rehabilitation game experience?*
- *RQ2: To what extent do the LA dynamics influence the performance of the patient tasks, and the level of physical exertion during the game?*
- *RQ3: What are age-related differences in user experiences and acceptance of the IVR-rehabilitation game, and how does aging impact patient engagement, motivation, and overall satisfaction with the rehabilitation process?*

To investigate these RQs, a study was conducted that integrated gamification techniques to increase patient motivation and engagement. The study followed a mixed-design approach, with LA and NLA as variables within subjects, and age groups (MAPs and OPs) as variables between subjects. In addition to evaluating motivation, gaming experience, and performance related to physical rehabilitation, the exertion of the patient was also measured. Qualitative data were also collected through interviews to explore the participants' feedback, specifically including their perceptions and acceptability of the LA dynamic: These were subsequently analyzed. This comparative approach was designed to explain how IVR-based gamification, with the LA dynamic, might influence short-term motivation, gaming experience, task performance, and physical exertion in the context of rehabilitation relevant to VTE prevention.

3 Methods

3.1 Game design approach

The user interface design of SPROUTFIT included careful consideration of elements such as color palette, structural layout, and font selection, guided by established design principles (Stone et al. 2005) and informed by collaboration with rehabilitation specialists. These considerations were important to ensure accessibility and ease of use for all participants, particularly elderly patients who may be experiencing cognitive decline. The chosen color scheme features a high contrast between a dark charcoal gray background

and white text for descriptive information, a decision made to maximize readability for users with varying levels of visual acuity. The accent colors, specifically yellow, were used to highlight key elements such as task and level indicators. The interface follows a clear modular structure in which key information is logically grouped within well-defined visual boundaries. The layout is optimized for a top-to-bottom information flow, facilitating sequential task progression and data acquisition, while minimizing the need for extensive navigation. Open Sans, a clean sans-serif font, was selected for its legibility. The primary text was rendered at a minimum size of 18 points, with key indicators using a larger scale to further ensure readability. To improve visual comfort and reduce visual clutter, we ensured sufficient padding (the empty space left) around text elements and adequate line spacing.

To examine the influence of LA for gamification on the rehabilitation experience for the prevention of VTE, SPROUTFIT was designed and developed based on the popular game *Farmville* (Balnaves et al. 2012), integrating LA into its game dynamic. The objective of SPROUTFIT was to involve patients in seed-planting activities, where the movements required for the planting were mapped to specific lower limb rehabilitation exercises aimed at preventing VTE. To increase intrinsic motivation (Naul and Liu 2020), a narrative game dynamic was also used: patients were in a world devastated by pests. Their task was to rebuild and restore the environment through planting activities. The game offered three different planting venues: a forest, a flower garden, or a farm (see Fig. 1), providing patients with options, as an engagement-improvement mechanism (Lomas et al. 2017). Each venue consisted of multiple stages and patients could gain or retain levels based on their performance — This was a LA core drive to serve as a motivation-enhancement design (Holmes et al. 2015).

SPROUTFIT consists of five stages, each corresponding to specific rehabilitation movements of the lower limbs. During each stage, patients were guided by a virtual instructional board displayed in the IVR environment (Fig. 2): This served as a reference for completing the assigned tasks. Real-time visual and auditory feedback were provided to patients about the accuracy of their movements in the virtual environment. The custom motion detection module (Sect. 3.2), using wireless communication, tracked and evaluated the movements. Correct movements were given scores and a corresponding auditory cue. In addition, a virtual animated guide of lower limb movements was kept visible (Fig. 2), guiding patients to replicate the correct movements to complete the tasks. Each stage had a 90-second time limit: The number of movements correctly completed within this period was used as an indicator of patients' game

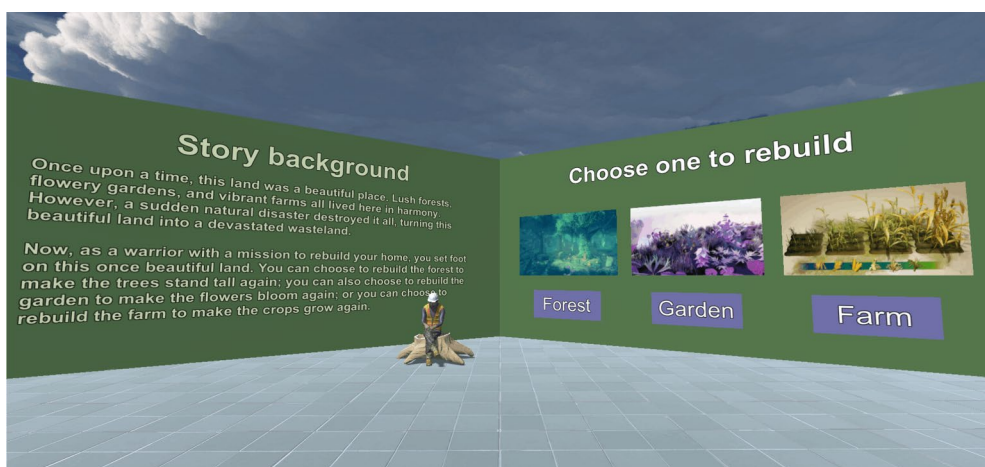


Fig. 1 Overview of SPROUTFIT design showing three planting venues: a forest, garden, and farm

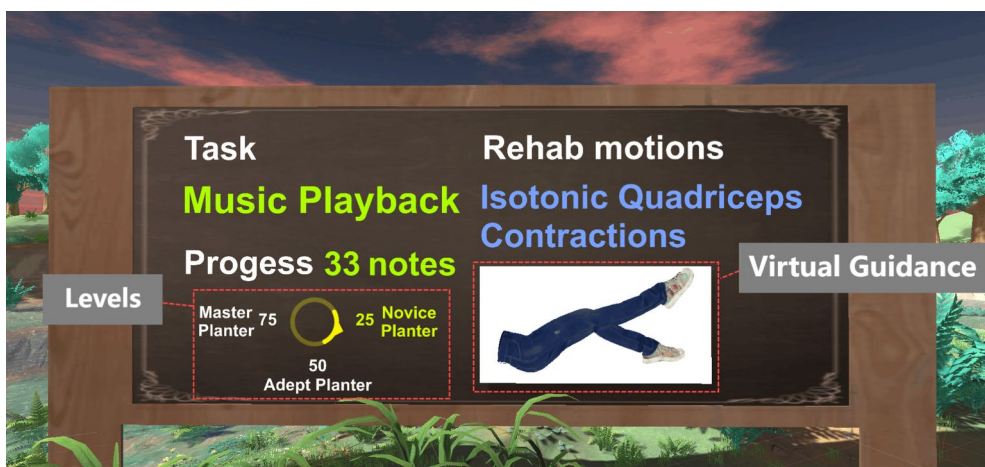


Fig. 2 An instructional board with virtual guidance displaying performance indicators for earning three different planter levels: Novice, Adept, and the highest, Master

scores (Each correctly executed movement contributed one game score).

The reward system includes the use of LA with three levels (Fig. 2): Patients were awarded a “Novice Planter” level at the beginning of each game stage. Earning higher scores allowed them to gain higher levels, aiming to maintain their motivation and involvement. They could gain the “Adept Planter” level after earning 50 scores (completing 50 successful movements), and the “Master Planter” level after earning 75 scores (achieving 75 successful movements) during each independent game stage. At the end of the game, a summary is presented indicating the level achieved at each stage, highlighting the general achievements of the patient. Each patient was asked to participate in two game sessions (within-subjects design) to address the RQs: LA-integrated SPROUTFIT (LA-SPROUTFIT) and non-LA-integrated SPROUTFIT (NLA-SPROUTFIT). In LA-SPROUTFIT game session, patients had to earn a minimum of 25 scores to retain the “Novice Planter” level, and to advance to the

next stage: This is consistent with the LA component of “loss”. The component of “avoidance” was implemented by requiring patients to restart a stage if they did not retain the “Novice Planter” level (by failing to achieve 25 scores): This was to motivate them to improve their performance. In contrast, patients in the NLA-SPROUTFIT game session could progress to the next stage regardless of performance, without applying LA components.

Five different lower limb movements for the prevention of VTE were designed in collaboration with clinical professionals at the First Affiliated Hospital of Ningbo University. These exercises were carefully designed through discussions between the research team and clinical professionals: They all followed well-established clinical practice guidelines (Xu et al. 2021). All movements were intended to be of low intensity, performed at 15 repetitions per minute, with a session range of 25 to 75 repetitions. The five SPROUTFIT stages were structured around these specific movements of the lower limbs, as shown in Table 1. Each stage included

Table 1 The lower limb movements for VTE prevention (based on clinical guidelines (Xu et al. 2021)) are associated with each game stage in sequential order. The movement frequency was set at 15 targets per minute, each game stage required at least 25 total movements, and each movement was of low intensity

Game stage (GS)	Lower limb movement	Description
GS1: seed collection	Ankle pump	Ankle pump involves lifting the front of the foot upward and pointing the toes downward
GS2: seed sowing	Ankle rotation	Rotating the foot in circular motions at the ankle joint
GS3: irrigation	Quadriceps set	Tightening the quadriceps without moving the knee joint
GS4: pest control	Straight leg raises	Lifting a straightened leg off the bed while lying down
GS5: music playback	Bed-unsupported knee bends	Contracting the quadriceps while allowing knee movement

unique seed-planting activities: These involve tasks of seed collection, sowing, irrigation, pest control, and music playback.

3.1.1 Seed collection (ankle dorsiflexion and plantarflexion)

In the first stage, patients performed ankle dorsiflexion and plantarflexion: each cycle of these movements triggered the generation of a virtual seed (Fig. 3a). When the patient's ankle returned to a relaxed position, the virtual seed generation button was reset to its initial state. The seed was only generated when the button was fully pressed: This required that the ankle joint achieve at least 60° of plantarflexion from its initial resting position to register as a completed movement. Partial button-presses did not produce seeds, which meant that no scores were earned. In LA-SPROUTFIT, patients had to generate at least 25 seeds to retain the “Novice Planter” level, and to progress to the next stage.

3.1.2 Seed sowing (ankle circles)

In the second stage, the patients performed ankle circles with both legs simultaneously to operate a seed-sowing machine (see Fig. 3b). Each seed was sown after a complete 360° counterclockwise rotation with the right ankle and a simultaneous 360° clockwise rotation with the left ankle. Partial rotations did not result in sowing, which means that no scores were earned. In LA-SPROUTFIT, at least 25 seeds had to be sown to retain the level, and to move on to the next stage.

3.1.3 Irrigation (isometric quadriceps contractions)

In the third stage, the patients controlled a virtual irrigation drone by performing isometric quadriceps contractions. The left foot controlled the drone's lateral movement and the right foot controlled the its longitudinal movement (Fig. 3c). Patients needed to continuously adjust their ankle movements, with at least 45° rotation, to navigate the drone over arid land areas: Failure to guide the drone to these areas resulted in no scores being earned. In LA-SPROUTFIT, patients had to irrigate 25 plots of land to progress.

3.1.4 Pest control (straight leg raises)

The fourth stage involved the use of straight leg raises to eliminate pests found in the crops. The virtual leg of the avatar served as a pest control spray (Fig. 3d): scores were awarded when the leg was raised to achieve a rotation within the range of 10° to 30°, pests were eliminated upon contact with the spray. In LA-SPROUTFIT, 25 successful sprays were required to keep “Novice Planter” level, and to proceed to the next stage.

3.1.5 Music playback (isotonic quadriceps contractions)

In the final stage, isotonic quadriceps contractions were used to directly control the tonearm of a long-playing record to play music, promoting seed growth (Fig. 3e). Patients' leg movements, requiring correct knee flexion within a range of 30° to 60° and a straight leg raise to at least 30° of hip flexion, had to be synchronized with on-screen guidance: Any incorrect movement halted the music. If the music stopped, then patients were unable to earn scores until they continued performing the movements correctly. In LA-SPROUTFIT, patients had to activate at least 25 music notes to retain the “Novice Planter” level.

3.2 Wearable sensor module design

SPROUTFIT was developed using Unity3D Editor v2022.3.20f1 (Unity Technologies 2024) and deployed on a Meta Quest 3 VR headset (Meta 2024): The headset had a 90 Hz refresh rate, a per-eye-resolution of 2064 × 2208 pixels, and a 110° field of view. The VR headset was connected to a laptop computer featuring an NVIDIA RTX 2000 Ada Generation GPU, an Intel i7-13800 H processor, and 32 GB of RAM.

The movements of the lower limbs of the patient were monitored using a motion-detection module (Wang et al. 2023). The motion-detection module comprises a BNO055 Inertial Measurement Unit (IMU) sensor (Bosch Sensortec), a Sseed Xiao Sense controller (Sseed Studio Bazaar),

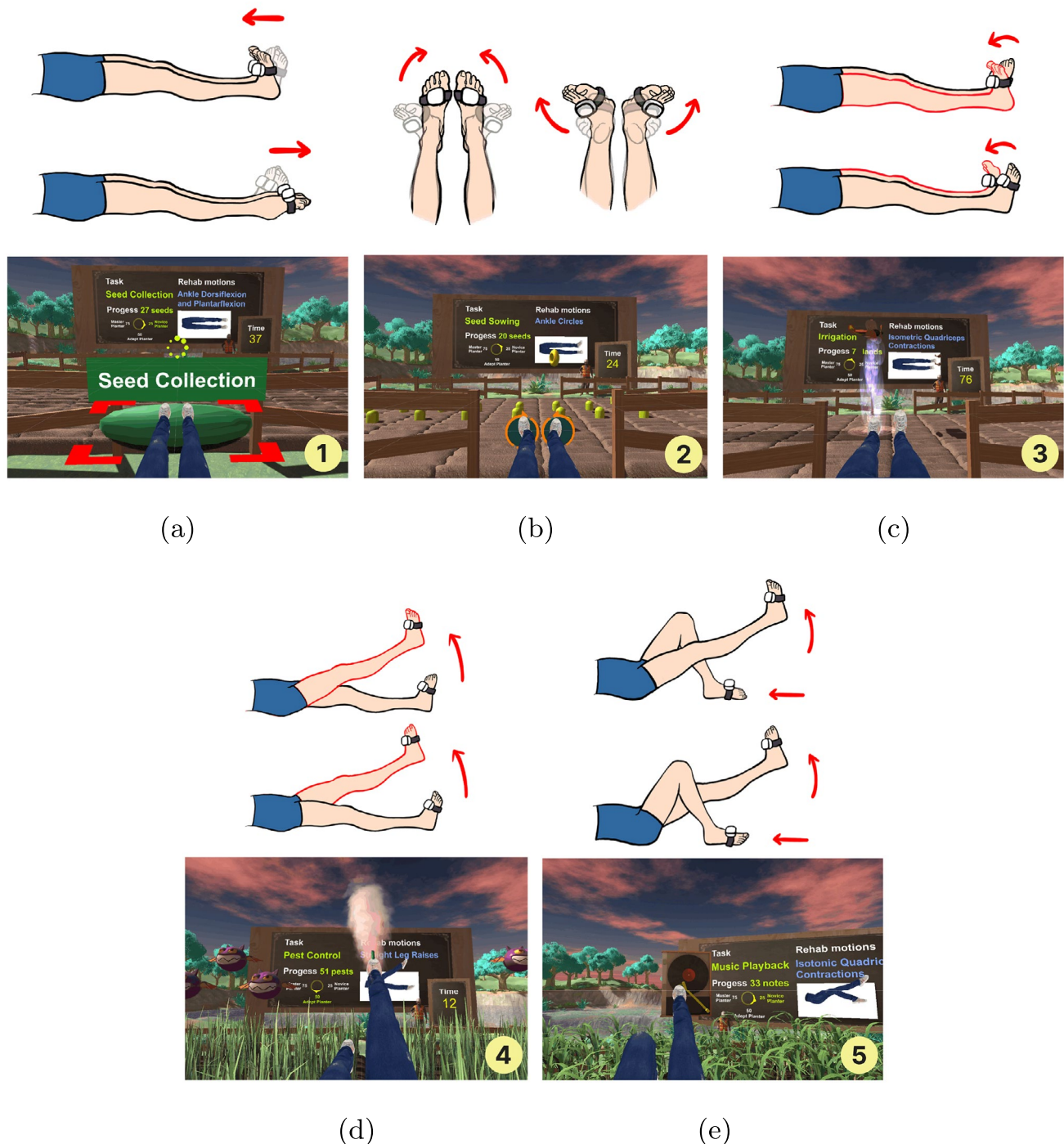


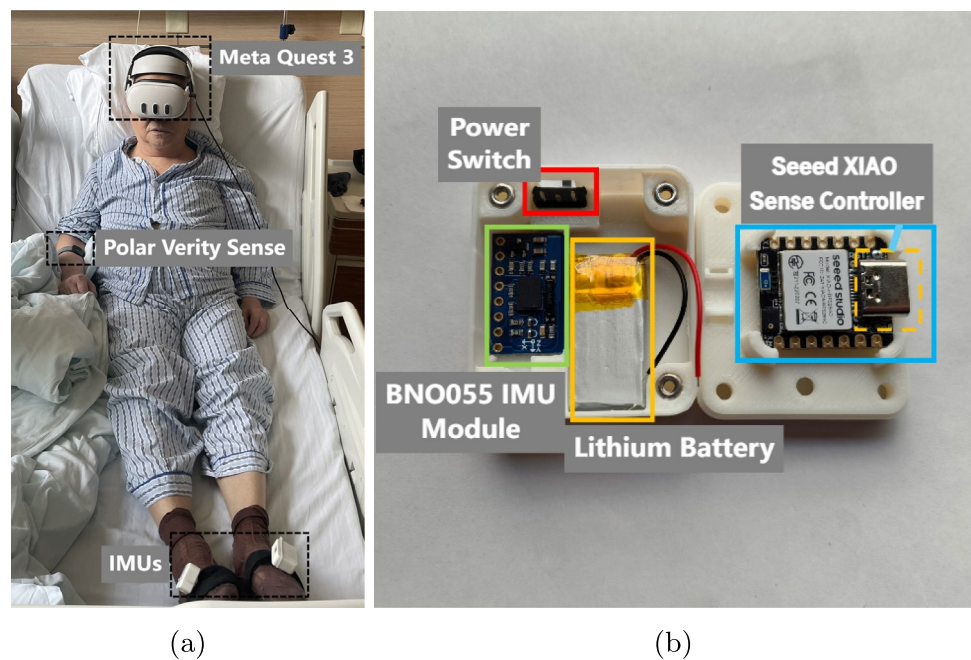
Fig. 3 The five stages of the IVR seed-planting game. **(a)** In Stage 1, the participants collected seeds by performing ankle dorsiflexion and plantarflexion. **(b)** In Stage 2, the participants performed circular ankle actions to sow seeds. **(c)** In Stage 3, the participants irrigated the land

a lithium battery, and a power switch (see Fig. 4b). The IMU sensor provided typical accuracies for various parameters, including yaw (heading) at $\pm 2.5^\circ$, pitch & roll at $\pm 1.0^\circ$, and gyro noise at approximately $\sim 0.014^\circ/s/\sqrt{Hz}$.

controlling a plane through isometric contractions of the quadriceps. **(d)** In Stage 4, the participants alternately raised their legs straight to spray and eliminate pests. **(e)** In Stage 5, the participants used isotonic quadriceps contractions to rotate a turntable and play music

The magnetometer accuracy was rated at $\pm 1.5 \mu T$ and the orientation (fused) at $1 - 2.5^\circ$ RMS. The IMU reported a calibration status (on a scale of 0–3, with 3 indicating full calibration) for each internal sensor (accelerometer, gyroscope, magnetometer) and the overall system. The IMU unit

Fig. 4 **a** Overview of a typical experimental setup where the participant is equipped with a Meta Quest 3 VR headset and a Polar Verity Sense armband to monitor heart rate and calories burned. **b** Lower limb movements are tracked using motion-detection modules attached to both feet, with data transmitted to a laptop connected to the VR headset



weighed 30.4 gs and had dimensions of 3.6 cm × 3.6 cm × 1 cm. It was attached to the center of the participant's foot using a flexible, soft strap to ensure comfort, and to minimize restriction of movement. The module collectively captured patient-movement data, which were subsequently transmitted to a laptop via Bluetooth for processing.

The laptop served as the central processing and response unit. Data processing involved extracting time-series data on orientation, angular velocity, and rotation. These data were used to determine the accuracy of the movement for the scoring and for progression in the game. The laptop terminal application performed two primary functions: mapping body-angle data to the local joint coordinate system; and extracting the current state of rehabilitation activity. Furthermore, the patient's heart rate (HR) and the calories burned were tracked using a Polar Verity Sense armband (produced by Polar Electro Malaysia (M) Sdn. Bhd) (Merigan et al. 2023).

Figure 4a shows an example of the experimental setup in the hospital, where a patient is shown using the VR headset and associated equipment during the experiment. In each session, the patient was guided to adjust their posture as shown in Fig. 4a. The game camera system was designed to be adaptive, allowing patients who may feel discomfort when seated to play the game comfortably in the reclined position. To ensure consistent tracking, and to prevent undue strain, the permissible range of motion in the hip joint was approximately 20° to 60° of flexion, adjusted according to the specific stage of the game and the individual patient's comfort level.

4 Experiment design

4.1 Participants

Two different age groups of hospitalized patients were recruited for the study: middle-aged patients (MAPs), aged 45–64 years; and older patients (OPs), aged 65 years and older. The recruitment process involved informing eligible patients about the purpose and procedures of the study, highlighting the voluntary nature of their participation, and the absence of compensation. The attending physicians also provided assurance that participation would not adversely affect the medical conditions of the patients.

Included patients satisfied the following inclusion criteria ((as confirmed by hospital rehabilitation specialists): (1) limited mobility, mainly bedridden after surgery, necessitating rehabilitation exercises to prevent VTE; (2) no reported history of motion sickness; (3) sufficient visual acuity and auditory function to effectively engage with the IVR environment; and (4) no reported use of medications known to significantly influence HR responses during exercise.

Patients were excluded if they had any of the following conditions ((as confirmed by hospital rehabilitation specialists): (1) any pre-existing medical condition to contraindicate participation in light physical activity or IVR-based exercises; (2) severe cognitive impairment that would impede their comprehension of game instructions, or their capacity to provide informed consent; (3) uncontrolled cardiovascular conditions; or (4) Severe discomfort or adverse reactions during initial exposure to IVR.

The majority of participants ($n = 21$) preferred a lying position when performed the exercises, while a smaller number ($n = 7$) chose to perform the exercises in a sitting position. Participants who opted for the sitting position were allowed to self-select whether to use back support, as experiments were carried out in a controlled hospital setting where patient comfort was prioritized in this study.

A total of 28 patients participated, comprising 14 MAPs (5 females, 9 males; $\mu_{age} = 55.36$, $\sigma_{age} = 7.68$, aged 45 to 64) and 14 OPs (6 females, 8 males; $\mu_{age} = 72.71$, $\sigma_{age} = 4.94$, aged 66 to 81). None of the patients had previous experience with IVR gaming.

4.2 Ethical consideration

This study was thoroughly reviewed and approved by the university's research ethics committee, according to the university's research conduct and ethical guidelines (Ethics 2024). The data were collected and securely stored on a password-protected laptop. Subsequently, the data underwent preprocessing before being stored on the university-managed secured cloud server, accessible only to members of the research team. The data only comprises user experience and anonymous physiological data: No information capable of revealing any individual participant's identity was recorded, thus ensuring anonymity within the dataset (e.g., participants were labeled as Subject 1, Subject 2, etc.). This entire process strictly adheres to ethical procedures.

Throughout the experimental sessions, the patients were closely monitored by nurses. The on-site nurses were authorized to terminate the experiment if they determined a patient could not continue, a protocol that ensured patient safety throughout the study. In this case, the recorded data would have been immediately deleted; however, no such instances requiring nurse-initiated termination occurred during the experimental period.

4.3 Experiment setup

Before the experiment, each patient received a 10-minute briefing on the purpose of the study. After agreeing to participate, the patients signed a consent form. The researcher and a hospital nurse then assisted each patient in wearing the VR headset and sensor modules on the body. Before the formal experiment began, the researcher ensured that all equipment

(including the laptop, the VR system, and the IMU sensors) was properly set up. It took an average of approximately five minutes to confirm that everything was fully ready to use.

Before starting the game, the patients participated in a brief tutorial to familiarize themselves with the interactions and mechanics of SPROUTFIT. During this time, patients were trained on how to use the controllers. The researcher ensured that they were comfortable with them before starting the game. Patients were explicitly informed that they could discontinue the experiment at any time if they felt discomfort. The current study involved a one-time session in which patients experienced SPROUTFIT. Both the nursing staff and the researcher were present, to ensure smooth operation and data collection. Had any patient wanted to withdraw, the situation would have been promptly managed; however, no requests to discontinue were made.

After completing a game session (whether it was LA-SPROUTFIT or NLA-SPROUTFIT), the patients had a break. The break period was at least five minutes, with the exact duration extended based on each patient's physical condition. The subsequent questionnaire completion time was approximately 15 min. The experiment then continued with the next round of game sessions. Upon completing both game sessions, patients were asked to complete the same questionnaire following a break. This was then followed by a 10-minute interview. An overview of the experiment is illustrated in Fig. 5.

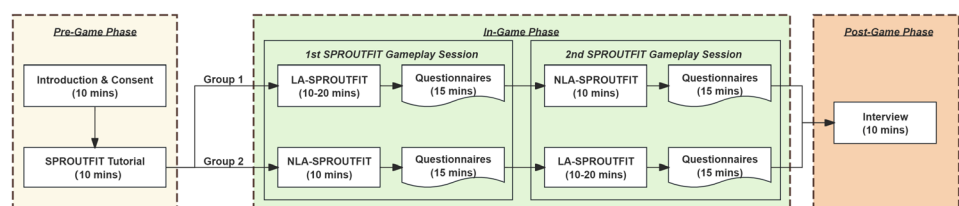
The study was conducted within the Coronary Care Unit (CCU) of The First Affiliated Hospital of Ningbo University (Hospital 2025). Data collection occurred between July and August 2024, with experimental sessions scheduled daily from 9:00 AM to 4:00 PM. Each patient in the study participated in two games sessions, with a total duration of two hours per patient.

4.4 Data measurements

This study used a cross-over design (Senn 2002), integrating a mixed methods approach to data analysis to conduct a comparative analysis of two key variables.

- Game Condition (GC: within-subjects variable): Each patient participated in both LA-SPROUTFIT and NLA-SPROUTFIT game sessions to investigate the impact on

Fig. 5 Overview of the experimental setup, covering the pre-game, in-game, and post-game phases



outcome variables (short-term motivation, game experience, game performance, and physical exertion), addressing RQ1 and RQ2. To mitigate any potential order effects, the session sequence was randomized using a Latin square counterbalance approach (Reese 1997).

- Age (between-subjects variable): To address RQ3 regarding differences in outcome variables, overall effectiveness, and satisfaction with the proposed SPROUTFIT between the two age groups (MAPs and OPs).

These measurement tools used included: The Gaming Motivation Scale (GAMS) (Lafrenière et al. 2012); the Game Experience Questionnaire (GEQ) (IJsselstein et al. 2013); the Borg Rating of Perceived Exertion (RPE) scale (Williams 2017); short interviews; evaluations of game performance; and physiological signals (such as HR and calories burned).

A two-way mixed analysis of variance (ANOVA) (Palant 2020) was used to examine the within-subjects and between-subjects variables. In instances where a significant interaction effect ($GC \times age$) was identified, a simple effects analysis was performed. The effect sizes were calculated to assess the magnitude of the observed effects. The sample size for this study was determined *a priori* based on a G*Power analysis (Verma and Verma 2020). Based on the following input parameters: effect size (f) = 0.3, error probability (α) = 0.05, power ($1 - \beta$ error probability) = 0.80, number of groups = 2, number of measurements = 2, correlation among repeated measures = 0.5 and nonsphericity correction $\epsilon = 1$, the required sample size was calculated as 24. Our actual sample size of 28 therefore meets and exceeds this criteria.

4.4.1 GAMS

The GAMS was used to assess various types of motivation that drive patient engagement with SPROUTFIT. This scale helped to identify the motivations that influence the devotion and progress of the patients in SPROUTFIT: It was based on a 7-point Likert scale, ranging from 0 (“Do Not Agree At All”) to 6 (“Very Strongly Agree”). GAMS measured motivations in several dimensions, including intrinsic motivation, integrated regulation, identified regulation, introjected regulation, external regulation, and amotivation.

4.4.2 GEQ

The GEQ provided valuable information on how patients perceived SPROUTFIT gaming experience: It followed a 5-point Likert scale, ranging from 0 (“Not At All”) to 4 (“Extremely”) and measured six dimensions (competence, sensory, flow, tension or annoyance, negative affect, and positive affect).

4.4.3 Borg RPE scale

The Borg RPE scale was used to assess the physical exertion felt by the patients during the experiment. This scale ranges from 6 (“No Exertion At All”) to 20 (“Maximal Exertion”). Patients reported their perceived exertion during or after completing specific rehabilitation tasks within SPROUTFIT. The scale allowed real-time feedback on how physically demanding the exercises — such as movements of the lower limbs or contractions of the quadriceps — felt while playing the game.

4.4.4 Interview

A short interview was conducted to assess the overall effectiveness, satisfaction, and acceptance of SPROUTFIT. This interview included two types of question: opinion-based questions (OQ) and evaluation-based questions (EQ). The OQs were designed to generate open feedback on the experiences of the participants, while the EQs focused on assessing usability, and identifying potential design issues. The questions included:

1. What are your thoughts on the requirement to complete 25 rehabilitation movements per game stage to maintain or gain a higher level, and progress to the next game stage? (EQ)
2. How would you describe your experience with SPROUTFIT? (OQ)
3. Were there any particular aspects of SPROUTFIT that caught your attention? (OQ)
4. Were there any aspects of SPROUTFIT that you found less engaging or enjoyable? (OQ)
5. Were there any tasks in SPROUTFIT that you found particularly difficult or unexpected? (EQ)
6. Did you encounter any parts of SPROUTFIT that were unclear or confusing? If so, could you describe them? (EQ)

To facilitate comprehensive feedback, the participants were not restricted in the length of their responses. Patients’ uncertainties during the interview often served as indicators of potential design flaws, prompting subsequent follow-up questions for further clarification (Drachen et al. 2018). The responses were recorded and subsequently analyzed using a general inductive approach (a qualitative method through coding and categorization) (Thomas 2006). This analytical process involved systematic data preparation, close reading of the transcripts, and feedback categorization. The primary analysis was conducted by a member of the research team and reviewed by other members of the team to ensure reliability and consistency.

4.4.5 Game score

The game score was determined by quantifying the number of correct lower limb movements performed by the patient during SPROUTFIT gameplay sessions. In LA-SPROUTFIT gameplay, the initial score achieved in each stage served as a key performance indicator.

4.4.6 Physiological data

Physiological data including HR and calories burned were analyzed. HR was measured as average beats per minute (bpm), while calories burned were measured as average kilocalories (kcal) per minute. These physiological indicators were used to assess the overall exertion of the patients in the rehabilitation process.

5 Results

The ANOVA results are summarized in Table 2. It is important to note that both the researcher and the nursing staff encouraged the patients to meet the recommended guideline of rehabilitation as much as possible. However, the

explicit minimum score requirement in LA-SPROUTFIT may have contributed to the increased focus and effort of the patient. This aligns with the core drive of LA, as the potential loss of the “Novice Planter” level may have motivated patients. In contrast, the absence of such a requirement in NLA-SPROUTFIT allowed patients to engage at their own pace, without the pressure of meeting a predefined target. The interpretation of the results below considers this key difference.

5.1 GAMS

The ANOVA analysis revealed a statistically significant difference in external regulation (motivation driven by external factors), with patients reporting higher mean scores in LA-SPROUTFIT compared with NLA-SPROUTFIT

($F(1, 26) = 21.198$, $p < .001$, $\eta_p^2 = .449$). No significant effects were observed for the other dimensions. Fig. 6 shows the mean GAMS scores for all participants, disaggregated by age group (MAPs and OPs), covering six dimensions (intrinsic motivation, integrated regulation, identified regulation, introjected regulation, external regulation, and amotivation).

Table 2 Results of the two-way mixed ANOVA for GAMS, GEQ, Borg RPE scale, game scores and physiological data

Measurement	Dimension	GC			Age			GC × Age		
		$F(1, 26)$	p	η_p^2	$F(1, 26)$	p	η_p^2	$F(1, 26)$	p	η_p^2
GAMS	Intrinsic motivation	2.296	.142	0.081	.333	.569	0.013	.930	.344	0.035
	Integrated regulation	.050	.824	0.002	.170	.683	0.006	.050	.824	0.002
	Identified regulation	3.228	.084	0.110	.039	.846	0.001	1.865	.184	0.067
	Introjected regulation	.379	.544	0.014	1.682	.206	0.061	.136	.715	0.005
	External regulation	21.198	< .001	0.449	1.062	.312	0.039	2.025	.167	0.072
	Amotivation	.571	.457	0.021	1.446	.670	0.053	.030	.863	0.001
GEQ	Competence	7.123	< .05	0.215	.032	.860	0.001	5.770	< .05	0.182
	Sensory	.129	.722	0.005	5.265	< .05	0.168	.129	.722	0.005
	Flow	13.346	< .01	0.339	.739	.398	0.028	1.351	.256	0.050
	Tension	2.113	.158	0.075	.221	.642	0.008	1.467	.237	0.053
	Challenge	11.742	< .01	0.311	.000	1.000	0.000	.455	.506	0.017
	Negative affect	.025	.875	0.001	.021	.885	0.001	.228	.637	0.009
	Positive affect	2.123	.157	0.076	3.957	.057	0.132	.860	.362	0.032
	—	14.093	< .001	0.352	.269	.608	0.010	.351	.558	0.013
Borg RPE scale (Exertion)	Stage 1	1.371	.252	0.050	1.766	.195	0.064	.469	.499	0.018
	Stage 2	12.447	< .01	0.324	.502	.485	0.019	1.336	.258	0.049
	Stage 3	3.551	.071	0.120	.138	.714	0.005	.034	.855	0.001
	Stage 4	12.461	< .01	0.324	.083	.776	0.003	.291	.594	0.011
	Stage 5	8.705	< .01	0.251	.030	.863	0.001	.025	.876	0.001
Physiological signals (Exertion)	HR	17.231	< .001	0.399	4.620	< .05	0.151	.082	.777	0.003
	Calories	38.378	< .001	0.596	.730	.401	0.027	.227	.638	0.008

*GC: Game Condition

*GAMS: Gaming Motivation Scale

*GEQ: Game Experience Questionnaire

*Borg RPE Scale: Borg Rating of Perceived Exertion Scale

Fig. 6 Comparison of mean Gaming Motivation Scale (GAMS) scores for LA-SPROUTFIT and NLA-SPROUTFIT for all participants, categorized by age group (MAPs and OPs), across six dimensions: intrinsic motivation, integrated regulation, identified regulation, introjected regulation, external regulation, and amotivation. Data are presented as mean $\pm$ standard deviation. Statistical significance levels are indicated as follows: *** $p < .001$

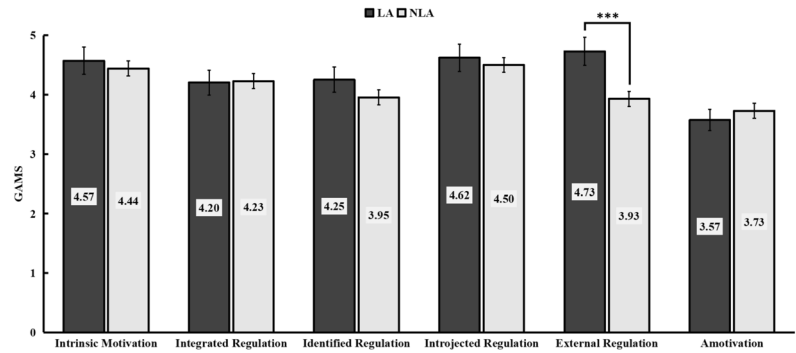
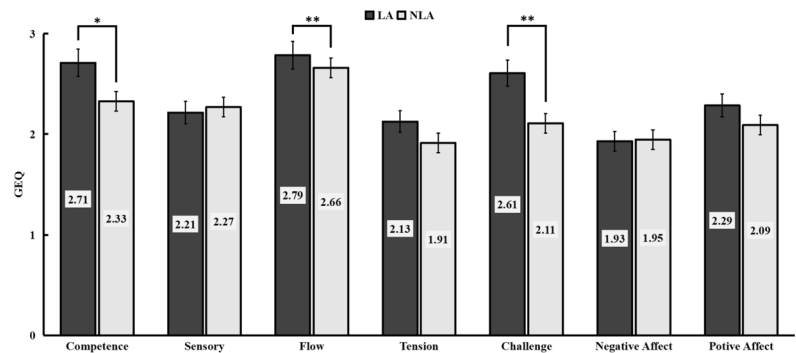


Fig. 7 Mean Game Experience Questionnaire (GEQ) scores for LA-SPROUTFIT and NLA-SPROUTFIT for all patients, categorized by age group (MAPs and OPs), across seven dimensions: competence, sensory, flow, tension, challenge, negative affect, and positive affect. Data are presented as mean $\pm$ standard deviation. Statistical significance levels are indicated as follows: * $p < .05$, ** $p < .01$



5.2 GEQ

The ANOVA analysis revealed a significant effect on competence ($F(1, 26) = 7.123$, $p < .05$, $\eta_p^2 = .215$), flow ($F(1, 26) = 13.346$, $p < .01$, $\eta_p^2 = .339$) and challenge ($F(1, 26) = 11.742$, $p < .01$, $\eta_p^2 = .311$) between LA-SPROUTFIT and NLA-SPROUTFIT. In addition, a significant effect of age was observed on the sensory dimension ($F(1, 26) = 5.265$, $p < .05$, $\eta_p^2 = .168$). These findings suggested that participants experienced higher levels of competence, flow and challenge with LA-SPROUTFIT sessions compared with NLA-SPROUTFIT. MAP participants reported greater sensory experiences compared to OP participants. Fig. 7 shows the mean self-reported ratings for all participants, disaggregated by age group (MAPs and OPs) in the GEQ, for competence, sensory, flow, tension, challenge, negative affect and positive affect.

Furthermore, a significant interaction effect (Rosnow and Rosenthal 1989) was identified in the competence dimension ($F(1, 26) = 5.770$, $p < .05$, $\eta_p^2 = .182$). A subsequent simple effects analysis (Pallant 2020) revealed that the MAP group rated their competence significantly higher with LA-SPROUTFIT than with NLA-SPROUTFIT

($F(1, 13) = 14.440$, $p < .01$, $\eta_p^2 = .526$). No other significant effects were identified.

5.3 Game score

The ANOVA results revealed a significant main effect of GC in Stage 2 ($F(1, 26) = 12.447$, $p < .01$, $\eta_p^2 = .324$), Stage 4 ($F(1, 26) = 12.461$, $p < .01$, $\eta_p^2 = .324$) and Stage 5 ($F(1, 26) = 8.705$, $p < .01$, $\eta_p^2 = .251$). These findings indicated that participants scored significantly higher in game Stages 2, 4, and 5 with LA-SPROUTFIT compared to NLA-SPROUTFIT. No other significant effects were observed. Fig. 8 presents the mean scores for all participants, disaggregated by age group (MAPs and OPs) across the five game stages.

5.4 Physiological data and borg RPE scale

The analysis revealed a significant main effect of GC on HR ($F(1, 26) = 17.231$, $p < .001$, $\eta_p^2 = .399$), calories burned ($F(1, 26) = 38.378$, $p < .001$, $\eta_p^2 = .596$) and the Borg RPE scale ($F(1, 26) = 14.093$, $p < .001$, $\eta_p^2 = .352$).

Fig. 8 Mean Game Score (GS) for LA-SPROUTFIT and NLA-SPROUTFIT for all patients, categorized by age group (MAPs and OPs), across the five game stages. Data are presented as mean $\pm$ standard deviation. Statistical significance levels are indicated as follows: * * $p < .01$

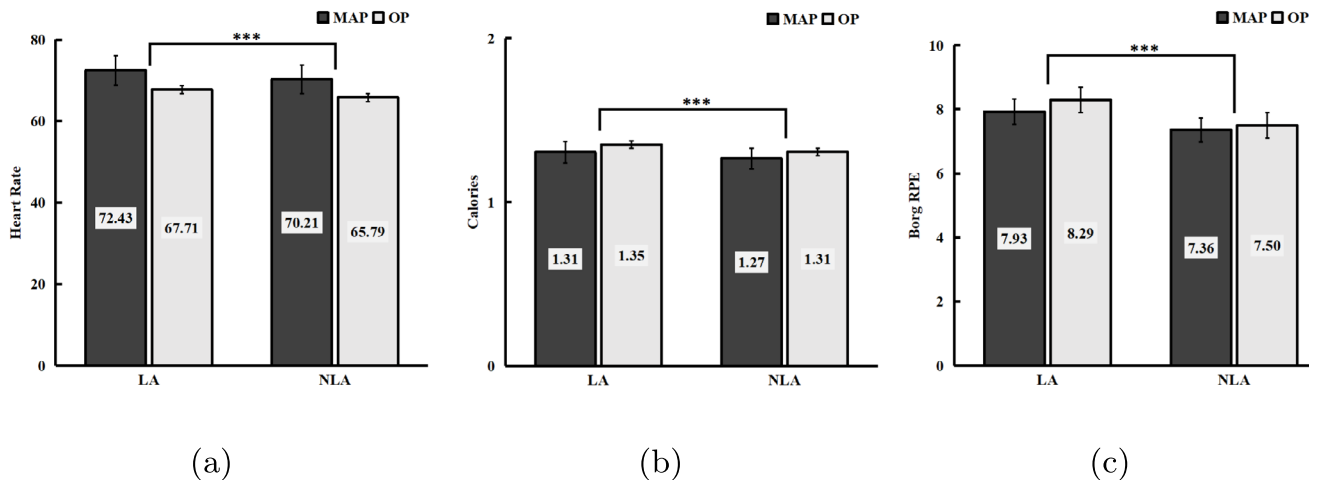
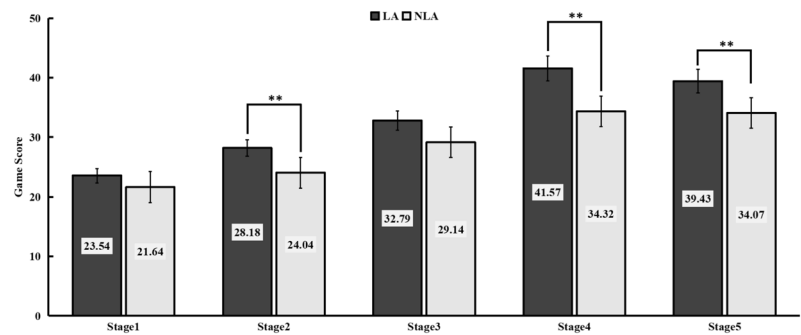


Fig. 9 Mean exertion values for LA-SPROUTFIT and NLA-SPROUTFIT for MAPs and OPs, over three dimensions: (a) heart rate (HR), (b) calories burned, and (c) Borg Rating of Perceived Exertion (RPE)

scale. Data are presented as mean $\pm$ standard deviation. Statistical significance levels are indicated as follows: * * * $p < .001$

Furthermore, a significant age effect was observed for HR ($F(1, 26) = 4.620$, $p < .05$, $\eta_p^2 = .151$). These findings indicate that the participants experienced higher HR, burned more calories, and reported higher Borg RPE scores with LA-SPROUTFIT than with NLA-SPROUTFIT. In addition, MAPs exhibited higher mean HR values compared to OP. No other significant effects were observed. Fig. 9 shows the mean HR, calories burned, and Borg RPE scale ratings for patients.

5.5 Interview

Four main categories from the analysis of the recorded transcripts: (1) overall gaming experience; (2) narrative gamification; (3) motivational levels and feedback; and (4) personalization and choice. Participants were labeled MAP1 to MAP14; and OP1 to OP14 for organizational clarity during the analysis.

Patients provided diverse responses to the proposed SPROUTFIT. Although some found the game intuitive from the outset, others required additional guidance to understand

the mechanics. However, in the end, all participants fully engaged in the game and provided informative feedback on their experiences. The game received generally high positive ratings, with several patients describing it as “enjoyable” (OP3, OP7), “motivating” (MAP5, OP11, OP12) and “rewarding” (MAP12). The inclusion of IVR technology was widely recognized as improving the experience, with most participants agreeing that the interactive environment contributed to a more dynamic rehabilitation process.

The narrative structure, which centered on seed-planting activities as part of a wider rehabilitation journey, received mostly positive responses (MAP4, MAP7, MAP9, OP2, OP5). MAP4 commented that the storyline made the exercises feel like “part of a meaningful mission”, while OP7 found it beneficial to a goal beyond just exercise, remarking that it made the game “more interesting”.

Five patients (MAP10, MAP13, OP1, OP2, OP10) found the inclusion of level mechanism particularly motivating, especially in LA-SPROUTFIT. However, MAP9 stated that this design occasionally introduced tension. Visual and auditory feedback triggered by correct movements was generally viewed as a positive feature. OP8 stated that the

feedback design “kept me on track and ensured that I was performing the exercises correctly”. Despite this, MAP11 suggested diversifying feedback methods, and OP14 proposed integrating audio commentary to enhance task comprehension and performance within the game.

Furthermore, MAP3 desired more choices within SPROUTFIT, suggesting that the narrative could be expanded to offer different outcomes based on patient decisions. OP12 acknowledged that while unfamiliar patients might initially find the game challenging, the LA design would become highly motivating once they became accustomed to it. OP5 also noted that although LA was well designed, introducing varying levels of difficulty could better accommodate the diverse needs of patients.

Patients did not report perceiving the seed-planting tasks involving lower-limb movements as counterintuitive. They indicated that the instructions were clear and informative, and that the virtual legs provided instant feedback to them that was meaningful. Participants reported that their focus on these virtual leg movements felt consistent and aligned with their reclined posture on the bed. Furthermore, they did not describe these lower-limb activities as being unnatural, or suggest that they should be performed with the hands. Individuals with similar previous rehabilitation experiences reported that SPROUTFIT diverted their attention from feelings of fatigue and boredom. These patients also reported that they made less effort to perform the prescribed lower-limb exercises compared to their previous rehabilitation experiences.

6 Discussion

In general, the statistical analyses showed that the inclusion of LA delivered higher mean score ratings in most dimensions related to motivation, user experiences, and exertion. Furthermore, the MAP group reported better experiences and acceptance of SPROUTFIT than the OP group.

6.1 LA vs NLA

This study provides evidence that the LA core drive, as integrated within our gamification framework, can effectively enhance engagement in exercises aimed at preventing VTE. Although the benefits of structured and progressive exercise routines for rehabilitation are well-established, including previous studies applying LA in various contexts (Xu et al. 2021), our single-session experiment provides further evidence that using LA can elicit more favorable responses compared to not using it: Patients demonstrated more satisfactory performance when completing lower limb movements and reported higher mean scores with LA.

6.1.1 Effect of LA on motivation and gaming experience

RQ1: How do LA dynamics influence patient short-term motivation and the perceived quality of the IVR-rehabilitation game experience?

The high scores for intrinsic motivation across both age groups indicate a high level of engagement with the rehabilitation tasks in SPROUTFIT, potentially indicating a pleasant user experience, which aligns with prior research on game-based rehabilitation (Callaghan et al. 2019). This appears to facilitate the satisfaction of psychological needs for autonomy, competence, and relatedness (Deci and Ryan 2008). Furthermore, the high scores for external regulation observed with LA-SPROUTFIT also indicate that the challenges inherent in the LA design may function as successful extrinsic motivation. Even though the findings show the potential for LA to enhance motivation and engagement within our study population, it is important to acknowledge that individual responses to LA may vary. Qualitative feedback, although limited in this study, revealed potential differences in how participants perceived the pressure associated with avoiding loss, with most participants reporting it as motivating.

Patients in LA-SPROUTFIT sessions reported significantly higher levels of competence, flow, and challenge compared to those in NLA-SPROUTFIT sessions. This finding indicates that patients felt more capable and confident in their ability to achieve their daily rehabilitation goals with LA: This could promote improved adherence. The observed increase in immersion (sense of presence in the virtual environment) may have contributed to reduced boredom and sustained attention to the game tasks. Furthermore, feedback from participants, including OP2’s suggestion regarding the inclusion of varying difficulty levels, highlights a potential avenue for future refinement to better accommodate the needs of different patients.

6.1.2 Effect of LA on game performance and physical exertion

RQ2: To what extent do the LA dynamics influence the performance of the patient tasks, and the level of physical exertion during the game?

Our analysis of game performance revealed a significant positive impact of LA. Patients in LA-SPROUTFIT sessions consistently achieved higher overall game scores in the rehabilitation tasks compared to those in the NLA-SPROUTFIT sessions. The observed higher exertion levels

with LA-SPROUTFIT — indicated by elevated mean heart rates, calories burned, and Borg RPE scale scores — suggest that patients exerted more effort during these tasks. Importantly, no fatigue or drowsiness were reported by patients with either LA or NLA. Furthermore, the observed positive correlation between the physiological measurements (elevated mean heart rates and calories burned) and the Borg RPE scale scores aligns with previous findings (Scherr et al. 2013).

Considering the limited mobility of the study participants (Sect. 4.1), SPROUTFIT appears to offer a potentially engaging and motivating activity for rehabilitation. However, although our findings are encouraging, more longitudinal studies are necessary to determine whether or not this observed increased short-term motivation and game experience translates into sustained adherence and improved long-term rehabilitation success in VTE prevention.

6.2 Effect of MAPs vs OPs in age-difference

RQ3: What are age-related differences in user experiences and acceptance of the IVR rehabilitation game, and how does aging impact patient engagement, motivation, and overall satisfaction with the rehabilitation process?

The absence of significant differences in game scores and motivation between the MAP and OP groups in this single session suggests that both age groups may perceive SPROUTFIT game as engaging for the specific low-exertion exercises performed in a reclined or sitting position. This aligns with the shared goal of rehabilitation for health recovery, and is consistent with previous research indicating the potential of gamification to enhance motivation across age groups (Baltaci et al. 2013). The qualitative feedback from patients MAP10 and OP9 further supports this observation, with both of them expressing consistent positive opinions regarding the rehabilitation goals for health. However, it is important to note that age-related comorbidities (multiple health conditions or diseases in one person, as highlighted by Divo et al. (2014)) could potentially influence engagement in more physically demanding rehabilitation settings — However, this question was outside the scope of the current study.

The significantly higher sensory and imaginative immersion (the feeling of actively involved in the virtual world's narrative and scenarios) reported by MAPs compared to OPs is consistent with the findings from Kimura et al. (2019). The difference may also be attributable to OPs having less prior exposure and familiarity with digital interfaces, coupled with established differences in digital

application usage patterns. This is further compounded by potential age-related declines in sensory processing among OP (Lindenberger and Ghisletta 2009), a factor consistent with the findings of Dermody et al. (2020). Therefore, when designing IVR-rehabilitation interventions for OP, it is important to prioritize interfaces and interactions that are intuitive, and that require minimal prior digital experience. Furthermore, adjusting for potential age-related sensory changes by offering adjustable visual and auditory feedback may enhance immersion and overall comfort. Furthermore, The observed higher HR in MAPs compared to OPs, despite similar reported motivation and engagement, suggests that age-related physiological differences in heart rate response to exercise could be a contributing factor (Fleg and Strait 2012).

6.3 Interview feedback

SPROUTFIT had a positive impact on patients' short-term motivation, as reported in the interviews: The majority of patients, across both age groups, reported enhanced engagement, and expressed a sense of enjoyment during game-based rehabilitation sessions. This suggests that the novelty of the IVR environment, and the initial gamified elements, effectively captured patient interest and encouraged active participation in the exercises.

However, while the observed short-term motivational effects are encouraging, a couple of patients (MAP12, OP7) expressed concerns regarding the potential for sustained engagement over longer periods. These patients, while acknowledging the initial novelty and enjoyment of the IVR experience, expressed that continued adherence to SPROUTFIT would require further design improvements. Their feedback indicates that initial novelty, while engaging, could diminish over time: They explicitly recommended incorporating game elements focused on sustained engagement to ensure continued participation.

6.4 Limitations and future work

This study faced several limitations that will inform future research directions. Even though sufficient visual acuity and auditory function were confirmed, we did not collect data regarding any cognitive decline or sensory impairments. Nor was a formal usability evaluation performed, particularly within the OP group. These factors could have influenced immersion and acceptance (willingness to continue using this technology). Future work should address this by including formal usability testing (e.g., heuristic evaluation and user testing (Nielsen 1995)): This will help to assess accessibility, and to understand how usability impacts satisfaction and efficiency across diverse patient profiles.

Table 3 The results from GAMS, GEQ, GS, and exertions

Measurement	Dimension	LA		NLA	
		MAP	OP	MAP	OP
GAMS	Intrinsic motivation	4.667	4.476	4.452	4.429
	Integrated regulation	4.167	4.238	4.214	4.238
	Identified regulation	4.381	4.119	3.857	4.048
	Introjected regulation	4.762	4.476	4.571	4.429
	External regulation	4.667	4.786	3.690	4.167
	Amotivation	3.643	3.500	3.762	3.690
GEQ	Competence	2.885	2.536	2.154	2.500
	Sensory	2.643	1.786	2.679	1.857
	Flow	2.643	2.929	2.571	2.750
	Tension	2.071	2.179	2.036	1.786
	Challenge	2.536	2.679	2.036	2.179
	Negative affect	1.893	1.964	1.964	1.929
	Positive affect	2.536	2.036	2.250	1.929
Borg RPE scale (exertion)	–	7.929	8.286	7.357	7.500
Game score	Stage 1	26.714	20.357	23.714	19.571
	Stage 2	28.643	27.714	25.857	22.214
	Stage 3	33.143	32.429	29.857	28.429
	Stage 4	41.500	41.643	35.357	33.286
	Stage 5	39.857	39.000	34.214	33.929
Physiological signals (exertion)	HR	72.429	67.714	70.214	65.786
	Calories	1.305	1.351	1.267	1.306

Regarding design aspects, although LA-SPROUTFIT showed promising impacts on short-term motivation, refinement is needed. Task comprehension proved challenging for some participants, especially in the OP group. This may have been due to differences in IVR-gaming experience. Integrating audio guidance could improve task clarity, particularly in patients with cognitive or sensory limitations. Furthermore, the static performance threshold (a non-adaptive, fixed criterion for game challenge) could be discouraging: A dynamic difficulty adjustment mechanism based on patient performance may be a potential solution. Future research should also investigate the impact of exercise position (seated and lying) during VR rehabilitation on VR gaming experiences in patients.

Building on the promising results for short-term motivation, and game experience, future work can focus on long-term experiments, and transition studies to home-based rehabilitation: This will help to assess long-term satisfaction, acceptance, and real-world feasibility. Alternative technologies should also be explored. Mixed reality (MR), for example, may provide a different experience for immersion and safety. Evaluating long-term motion sickness should also be under taken. Importantly, future work should

verify the direct clinical utility of SPROUTFIT by assessing its effect on objective VTE clinical measures (such as venous Doppler ultrasound scans, Leg circumference/swelling reduction).

7 Conclusions

This paper has introduced SPROUTFIT, an IVR game that integrates seed-planting gamification with LA, to encourage lower limb movements for the prevention of VTE. A mixed-design approach assessed the effects of LA on patient motivation, gaming experience, rehabilitation success, and physical effort. The results show that LA-SPROUTFIT significantly increased extrinsic motivation, gaming experience regarding competence, flow, and challenge, resulting in improved short-term motivation in rehabilitation, and higher levels of physical exertion, for both MAPs and OPs. Importantly, MAPs reported a stronger sense of competence in the LA game condition, while OP encountered difficulties fully engaging with the sensory aspects of the IVR environment. The analysis emphasizes the substantial positive influence of LA on SPROUTFIT, while also highlighting a need to introduce a dynamic difficulty mechanism adapted to different groups of patients (based on their abilities, and age). This study provides valuable information, demonstrating that LA can be a game dynamic to improve patient motivation, satisfaction, and acceptance, particularly within the MAP and OP populations.

Appendix a overall data results

Table 3 presents data from Figs. 6, 7, 8 and 9 for a comparative overview.

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Data availability The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no conflict of interest.

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Consent for publication All authors have read and approved the final manuscript for publication. The authors confirm that all participants provided their informed consent for publication.

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